

Smart Recipe Ontology (SRO)

Ontology Requirements Specification Document - Version 2

Field	Value
Project	Smart Recipe Ontology (SRO)
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Course deliverable	Ontology Development Project - Phase 2
Ontology version	2.0
Specification version	2.0
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GitHub repository	<GITHUB_REPOSITORY_URL>
Widoco documentation	<WIDOCO_DOCUMENTATION_URL>

1. Purpose

The purpose of the Smart Recipe Ontology (SRO) is to provide a structured, machine-readable knowledge model for cooking recipes. Version 2 represents recipes, ingredients, ingredient quantities, preparation steps, nutrition information, dietary suitability, allergens, cuisine, kitchen utilities, provenance, and data acquisition metadata. It supports semantic search and future recommendation systems based on ingredients, diet, calorie values, cooking time, preparation workflow, and user preferences.

2. Scope

The ontology focuses on recipe structure and recipe-oriented querying. It references ingredients, utilities, diet categories, units, and nutrition values only as needed to answer recipe competency questions. It is not intended to be a complete food science ontology, medical nutrition ontology, or full kitchen-equipment ontology. Version 2 adds ontology population and data provenance support so that recipe records can be acquired from datasets, APIs, or documents and mapped into RDF/OWL.

3. Implementation Language

OWL 2 / RDF, implemented in Turtle and RDF/XML serializations. Main file: ontology/smart-recipe-ontology-v2.ttl. Protege-compatible RDF/XML file: ontology/smart-recipe-ontology-v2.owl.

4. Intended End-Users

- Home cooks searching recipes by ingredients, time, cuisine, or diet.
- Diet planners and users checking calorie or allergen-related constraints.
- Recipe recommendation systems and chatbot/QA systems.
- Developers building recipe knowledge graphs or semantic applications.
- Students and ontology engineers validating competency questions with SPARQL.

5. Intended Uses

- Search recipes based on available ingredients.
- Filter recipes by dietary preference such as vegan, vegetarian, gluten-free, or high-protein.
- Retrieve ingredient quantities, units, and preparation states for a recipe.

- Retrieve ordered preparation steps and required kitchen utilities.
- Analyze calories, macro-nutrients, servings, cooking time, and preparation time.
- Populate a recipe knowledge graph from recipe datasets, APIs, or text documents.
- Support future personalization by matching recipes with user preferences.

6. Ontology Requirements

6.1 Non-functional requirements

ID	Requirement
NFR1	The ontology shall support English labels and comments.
NFR2	The ontology shall follow OWL/RDF standards and be editable in Protege.
NFR3	The ontology shall be reusable and extendable through modular classes and properties.
NFR4	The ontology shall include provenance for acquired recipe records.
NFR5	The ontology shall avoid inventing missing nutrition or ingredient values during automatic extraction.
NFR6	The ontology shall be documented through Widoco or a Widoco-ready documentation workflow.
NFR7	The ontology shall be version-controlled in GitHub with semantic version v2.0.

6.2 Functional requirements and competency questions

CQ	Question	Required ontology elements
CQ1	Which recipes are vegetarian or vegan?	sro:Recipe, sro:VegetarianRecipe, sro:VeganRecipe, sro:suitableForDiet, sro:DietType
CQ2	Which recipes use a specific ingredient?	sro:Recipe, sro:IngredientUsage, sro:usesIngredient, sro:Ingredient
CQ3	What ingredients are used in a recipe?	sro:hasIngredientUsage, sro:usesIngredient, sro:quantityValue, sro:unitLabel
CQ4	How many calories does a recipe contain?	sro:hasNutritionProfile, sro:caloriesPerServing
CQ5	How long does it take to cook or prepare a recipe?	sro:prepTimeMinutes, sro:cookTimeMinutes, sro:totalTimeMinutes
CQ6	Which kitchen utilities are required for a recipe?	sro:hasPreparationStep, sro:requiresUtility, sro:KitchenUtility
CQ7	What are the sequential steps to prepare a dish?	sro:PreparationStep, sro:stepNumber, sro:instructionText, sro:nextStep, sro:precedesStep
CQ8	Which recipes are suitable for a gluten-free diet?	sro:GlutenFreeRecipe, sro:GlutenFreeDiet, sro:suitableForDiet
CQ9	Which recipes can be prepared under a time limit?	sro:totalTimeMinutes
CQ10	What source was a recipe derived from?	sro:DataSource, sro:derivedFromSource, sro:sourceUrl
CQ11	Which recipes match a user preference profile?	sro:UserPreference, sro:matchesPreference, diet/time/calorie filters

7. Pre-Glossary of Terms

Category	Terms
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Terms from competency questions	Recipe, ingredient, calorie, cooking time, preparation time, diet type, kitchen utility, gluten-free diet, user preference, data source.
Terms from expected answers	Vegan recipe, vegetarian recipe, gluten-free recipe, ingredient list, preparation step, nutrition profile, cuisine, course category.
Objects/classes	Recipe, Ingredient, IngredientUsage, PreparationStep, NutritionProfile, DietType, Allergen, Utility, Unit, Cuisine, CourseCategory, CookingMethod, DataSource, UserPreference.

8. Data Acquisition

Version 2 includes an explicit data acquisition model and a sample dataset for testing. The acquisition process supports APIs, public datasets, and manually validated documents.

Data source	Role	Mapping target
Manual validation CSV	Trusted sample for testing and demonstration.	Recipe, IngredientUsage, PreparationStep, NutritionProfile, DietType, DataSource.
TheMealDB API	Recipe JSON source for names, categories, areas/cuisines, instructions, ingredients, and measures.	Recipe, Cuisine, CourseCategory, PreparationStep, IngredientUsage.
RecipeNLG dataset	Large recipe text dataset for future extraction experiments.	Recipe, Ingredient, PreparationStep.
USDA FoodData Central API	Nutrition enrichment for ingredient-level calories and nutrients.	Ingredient, NutritionProfile, fdclid.

Preprocessing steps: deduplicate records; filter English recipes; parse ingredient lines; normalize units; segment preparation steps; map diet/allergen tags; enrich nutrition only from trusted data; validate triples and record provenance/confidence.

9. Existing Ontologies and Extension Strategy

Used/extended vocabulary	Usage
Schema.org Recipe	SRO aligns sro:Recipe with schema:Recipe and references recipeIngredient, recipeInstructions, recipeCuisine, recipeCategory, cookTime, prepTime, and suitableForDiet.
Schema.org NutritionInformation	SRO aligns sro:NutritionProfile with nutrition information and calories.
QUDT	SRO aligns unit individuals such as Gram and Milliliter with QUDT unit identifiers.
Dublin Core Terms / PROV style metadata	SRO uses creator, title, created/modified date, version, and data source provenance concepts.

SRO does not import a large food ontology such as FoodOn in Version 2 because the project scope is recipe retrieval, not full biological or chemical food classification. Ingredient categories are modeled only as needed for diet/allergen filtering.

10. Ontology Expansion Methodology

The Version 2 extension follows a hybrid METHONTOLOGY and ODP/MOMo-inspired modular approach.

Methodology element	Application to SRO
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METHONTOLOGY - specification	Updated purpose, scope, users, uses, and competency questions.
METHONTOLOGY - conceptualization	Created modules for recipe structure, ingredient usage, preparation process, nutrition, diet/allergen, provenance, and user preference.
METHONTOLOGY - formalization/implementation	Implemented classes, object properties, datatype properties, labels, comments, and example individuals in OWL/RDF.
METHONTOLOGY - evaluation	Validated competency questions using SPARQL queries.
ODP - n-ary relation pattern	IngredientUsage captures ingredient + quantity + unit + preparationState + confidence.
ODP - sequence pattern	PreparationStep uses stepNumber, nextStep, and precedesStep.
MOMo-inspired modularity	SRO is divided into small modules that can guide future LLM-based ontology population.

11. Version 2 Change Log

Change ID	Change from v1 to v2
CH1	Added explicit data acquisition section and mapping from raw recipe fields to ontology concepts.
CH2	Added research integration based on ontology population using LLMs.
CH3	Added mandatory paper review adaptation through UserPreference and future personalization.
CH4	Added IngredientUsage class for quantities, units, preparation states, and extraction confidence.
CH5	Added NutritionProfile with calories, protein, fat, and carbohydrate properties.
CH6	Added DietType, Allergen, Cuisine, CourseCategory, CookingMethod, DataSource, and UserPreference modules.
CH7	Added SPARQL competency queries and query results.
CH8	Added GitHub versioning plan and Widoco documentation instructions.

12. GitHub and Documentation

GitHub repository: <GITHUB_REPOSITORY_URL>

Widoco documentation: <WIDOCO_DOCUMENTATION_URL>

Repository update should include ontology files, data sample, queries, reports, documentation, and a v2.0 tag.

13. Files Associated with This Specification

- ontology/smart-recipe-ontology-v2.ttl
- ontology/smart-recipe-ontology-v2.owl
- data/sample_recipe_data_phase2.csv
- queries/competency_queries.rq
- queries/query_results.txt
- reports/SRO_Phase2_Report.docx
- reports/SRO_Specification_Document_v2.docx
- README_GitHub_Widoco.md

14. References

1. Gultekin, A., Sepik, E., and Tanriver, S. Smart Recipe Ontology Phase 1 ORSD, 2026.
2. Norouzi, S. S., Barua, A., Christou, A., Gautam, N., Eells, A., Hitzler, P., and Shimizu, C. Ontology Population using LLMs, 2024.
3. Fernandez, F. M. H., Venkata Ramana, T., Shabana, M., Kannagi, V., and Nalini, M. Personalized ontology and deep training tree-based optimal gated recurrent unit-recurrent neural network for prediction of students behavior, 2023.
4. Schema.org Recipe and NutritionInformation documentation.
5. QUDT ontology documentation.
6. USDA FoodData Central API Guide.
7. TheMealDB API documentation.
8. Widoco documentation.